Sentiment Analysis With Self-MM Under Missing and Noisy Modalities

Agreement Between Unimodal Predictions as a Fusion Weight

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Repository: https://github.com/ilaydagencer/yzm5110-robust-selfmm
Submitted version (tag v1.0-submission): https://github.com/ilaydagencer/yzm5110-robust-selfmm/tree/v1.0-submission
Working branch: yzm5110-selfmm

Abstract

This study reproduces Self-MM [1], which uses text, audio and video, on CMU-MOSI with settings adapted to a 4 GB GPU and with five training seeds, and tests whether the agreement between its unimodal predictions can serve as a fusion weight when a modality is missing or noisy. The measured clean test MAE is 0.7240 ± 0.0146 (0.713 in the paper); Non0 accuracy is 2.08 percentage points below the paper's. Agreement weighting without extra parameters (τ=1) gave no meaningful improvement in any condition: it increased the error in every audio/visual condition and every seed (+0.118 [+0.091; +0.148] on the clean test set) and doubled inference time. The small validation gain of audio/visual dropout during training did not carry over to the test set. To separate the possible causes, a diagnostic protocol, registered before the results were seen, was run; all six decisions matched the expectation and replicated on test. The harm comes from rescaling the dominant text representation; the agreement signal is no better than randomly shuffled weights; and because the model hardly uses audio and video, even oracle weights cannot improve results when these modalities are corrupted. When text is corrupted, the error rises sharply (+0.675 when all text is removed), yet the rule increases the text weight instead of lowering it. This rule therefore cannot be used as a reliability signal, and on MOSI, insensitivity to audio/visual corruption should not be read as robustness.

1 RESEARCH QUESTION, SCOPE, MOTIVATION AND CONTRIBUTIONS

Research questions: (Q1) Can the agreement between Self-MM's unimodal predictions be used to set the fusion contributions when audio or video is missing or corrupted; does this rule preserve performance on clean input while reducing the error on corrupted input? (Q2) Does independent audio/visual dropout during training change this behavior? (Q3) If the rule does not work, is the reason the signal itself, the way the weight is applied, or that the corrupted modalities do not matter to the model?

Motivation: Multimodal models encounter missing or faulty sensors in real use. If a reliability cue could be obtained from the unimodal predictions the model already produces, without training an extra confidence module, it would be a cheap route to robustness. Because Self-MM has separate text, audio and video heads, it is a suitable base for testing such a signal; recent work such as LNLN and HME instead handles reliability with learned modules [5,6].

The scope is the pre-extracted unaligned CMU-MOSI features, BERT-based Self-MM, five training seeds and eleven conditions: eight conditions applied to audio/video (K0–K7) and text corruption added for Q3 (K8–K10). Raw sensor noise, other languages or datasets, training a new fusion network and retraining with the weighting are out of scope. Besides the sentiment direction, the model predicts sentiment intensity between −3 and +3; MAE is therefore the primary metric.

Contributions: (i) a traceable five-seed reproduction of Self-MM adapted to a 4 GB GPU, with an explicit list of deviations from the paper; (ii) a controlled test of a parameter-free agreement weighting rule and of training-time dropout with an A/B/C/D design, under shared corrupted inputs, reporting quality together with cost; (iii) a diagnostic protocol, committed before the results were seen, that separates the causes of the negative result (τ sweep, text-fixed, shuffled and oracle controls, text corruption), and its replication on test; (iv) a repository that reproduces the tables and figures without a GPU. A negative result does not invalidate the research question. The report does not claim that Self-MM robustness tests are done here for the first time, or that the method is the best in the field; LNLN has already evaluated Self-MM on missing data [5].

2 LITERATURE SEARCH APPROACH AND CRITICAL SYNTHESIS

Search and Selection

On 18 October 2026, three targeted searches were run with the Exa search engine: for the original Self-MM paper, for the full text of TFR-Net, and for multimodal sentiment analysis with missing or noisy modalities between 18 October 2023 and 18 October 2026. The searches returned 20 results in total; this does not mean that 20 papers were read in detail. Duplicates, sources that only provide abstracts and off-topic results were removed. To position the base methods correctly, LMF and MulT were traced from Self-MM's references and their official ACL Anthology pages. The method, experiment and limitation sections of the six selected works were studied from the primary full texts. Code availability was recorded from the official repositories as far as it could be verified.

The selection criteria were a direct relation to MOSI, coverage of different method families, relevance to the research question and access to the primary full text. Accordingly, LMF represents the computationally efficient base approach, MulT unaligned interaction, Self-MM multi-task representation learning, TFR-Net reconstruction of missing modalities, and LNLN and HME more recent approaches to missingness and reliability.

Common Ground and Differences Between the Methods

LMF models interactions with less computation; MulT learns dependencies between time series. Having a strong fusion mechanism does not mean that either approach estimates how reliable a missing modality is. Self-MM instead produces separate auxiliary targets so that different modalities are not forced onto a single sentiment label. The purpose of this mechanism is to learn modality-specific information, not to estimate confidence. [1–3]

TFR-Net and LNLN use additional learning signals about the missing information. HME combines information across samples with learned uncertainty. Our rule instead uses the agreement of the three predictions, without a separately trained confidence module. This keeps the implementation simpler, but it assumes that the three predictions are on the same scale and comparable in reliability. If two wrong auxiliary predictions are close to each other, they can suppress the correct modality. [4–6]

An important scope correction: LNLN has already evaluated Self-MM under missing data. The testable gap here is whether the chosen unimodal agreement rule helps under this implementation and protocol. In addition, LNLN corrupts the text, uses different features and reports separate checkpoints for regression and classification, so its results cannot be ranked directly against our single-checkpoint tables. [5]

Literature Comparison Matrix

Paper	Question and contribution	Modalities and assumptions	Mechanism and objective	Data and protocol	Baselines and evidence	Resources and reproduction	Limitations and links
[1] Self-MM / Yu et al. / AAAI 2021	Generates modality-specific targets from a single multimodal label; reduces the need for hand-made unimodal labels.	Text, audio, video. A joint sentiment label in training; three inputs at inference. The generated targets are not confidence scores.	BERT and LSTMs; concatenation; ULGM with class centers; momentum target updates; joint and weighted unimodal L1.	MOSI, MOSEI, SIMS; pre-extracted features. MOSI 1284/229/686; the BERT version uses unaligned inputs.	LMF, MulT, MISA, MAG-BERT; task-removal experiments. Some competitor scores are taken from other papers; feature differences matter.	Official Self-MM code and an MMSA implementation exist. Mean of five runs. This project used MMSA; it is not the same as the authors' code.	Learning modality-specific separation is not learning corruption reliability. This project tests that transfer assumption.
[2] LMF / Liu et al. / ACL 2018	Reduces the size and cost of tensor fusion with a low-rank factorization.	Text, audio and video representations; a task label. The main evaluation relies on complete, word-aligned inputs.	Modality-specific low-rank factors are multiplied and summed; no explicit large outer-product tensor is built.	MOSI, POM, IEMOCAP; GloVe text; P2FA alignment; speaker-independent splits.	Direct fusion comparison with TFN; rank sensitivity, parameter and speed analysis. A raw score comparison with a BERT-based work is not like for like.	Official code link; Quadro K4200; training and inference speed reported. No matched five-seed experiment in this project.	Efficient fusion does not guarantee robustness to a missing modality. Gives a basis for choosing the resource metric.
[3] MulT / Tsai et al. / ACL	Handles different sampling rates	Text, audio and video sequences; a sentiment/task	Directional cross-modal attention matches the	MOSI, MOSEI, IEMOCAP; aligned and	Early/late fusion transformers and earlier models;	Official code; a single GTX 1080 Ti.	Success at alignment is not evidence

Paper	Question and contribution	Modalities and assumptions	Mechanism and objective	Data and protocol	Baselines and evidence	Resources and reproduction	Limitations and links
2019	and long-range dependencies without explicit alignment.	label. Misalignment is not the same problem as a missing modality.	queries of one modality with the keys and values of another; sequential fusion is learned.	unaligned protocols. Language features and preprocessing differ from the Self-MM BERT setup.	cross-attention and modality-direction controls.	Computation scales with sequence length; not retrained here.	of confidence calibration. An alternative interaction mechanism to Self-MM's simple fusion.
[4] TFR-Net / Yuan et al. / ACM MM 2021	Aims to reconstruct the semantics of randomly missing features in unaligned sequences.	Three modalities; complete features and missing positions are used for reconstruction in training. Inputs are corrupted at inference.	Intra- and cross-modal attention, a feature generator, a CNN gate and fusion; sentiment L1 and SmoothL1 reconstruction loss.	MOSI and SIMS; BERT, MOSI audio 5 / video 20 dimensions. UNK for text, zeros for the others; the same missing rate in train/valid/test.	TFN, MulT, MISA; module removal; missing-rate curves. The effect of the class distribution at high missing rates on SIMS is discussed.	Official code; tuning on validation, three seeds. The main text studied does not state the GPU model; no checkpoint was used.	Needs complete training targets. A direct score comparison with our fixed-checkpoint protocol is not possible: the main conditions (K0–K7) corrupt only audio/video, and text corruption (K8–K10) is a diagnostic that is never seen in training.

Paper	Question and contribution	Modalities and assumptions	Mechanism and objective	Data and protocol	Baselines and evidence	Resources and reproduction	Limitations and links
[5] LNLN / Zhang et al. / NeurIPS 2024	Aims at robustness to missing data by correcting the dominant language modality.	Three modalities; the input before corruption and the degree of missingness in training; completeness estimated from the missing data at inference.	Language completeness estimation, a proxy representation from auxiliary modalities, adversarial learning, reconstruction and sentiment losses.	MOSI, MOSEI, SIMS; missing rates 0–0.9. Feature extraction and the corruption of text also differ from the local protocol.	MMSA competitors including Self-MM, and module/loss removal. Reports separate best checkpoints for regression and classification.	Tesla A40, PyTorch 2.2.1; seeds 1111–1113, batch 64. Hyperparameter table and code link available.	Not always the best under complete modality loss. A Self-MM missing-data comparison already exists; a first-time claim cannot be made.
[6] HME / Zhuang et al. / NeurIPS 2025	Enriches the available representation with cues from other samples instead of explicitly generating the missing modality.	The available subset of the three modalities; access to other samples in the batch. Under the random protocol the lost modality is not a reconstruction target.	Cross-sample selection, learned prompts, a variational information bottleneck and uncertainty-aware fusion; task loss.	MOSI, MOSEI, IEMOCAP; fixed and random whole-modality loss. At least one modality remains; our time-step mask is different.	GCNet, MPLMM, DiCMoR, IMDer, LNLN; component removal and examples of harmful enrichment.	Settings in the paper's Appendix A, five repetitions, a 3090 GPU and a resource analysis. Code/checkpoint availability was not independently verified here.	Similar samples can reinforce the wrong sentiment; the batch context matters. Learned uncertainty differs from agreement between predictions.

3 TECHNICAL DESCRIPTION OF THE CHOSEN METHOD

Self-MM From Input to Prediction

In the MMSA implementation, text is encoded with BERT-base-uncased and the first token representation is taken. Audio and video pass through LSTM encoders that use the true sequence lengths. In the local settings the text, audio and video representations have 768, 16 and 32 dimensions. The concatenated 816-dimensional vector enters the 128-dimensional first fusion layer and the following prediction layers. Separate T/A/V heads produce unimodal predictions from the same encoders. The BERT weights are updated during training. [1,7; local model SELF_MM.py, forward]

There are four tasks: M is the multimodal prediction; T text; A audio; V video. These task letters are distinct from the names of the experimental methods A/B/C/D. Zeroing a modality's input is not the same as zeroing its representation: the biases of the LSTM and linear layers can produce a non-zero representation or prediction from a zero input.

Self-Supervised Unimodal Targets

At the start, all three auxiliary targets equal the observed multimodal label. For each task, the representation centers of the positive and negative samples are kept. Because excludeZero=True in the local code, zero labels are not included in the positive group of the centers. The relative difference between a sample's distances to the two centers shifts the unimodal target away from the multimodal target. This does not create independent human-provided audio/video labels; it produces an auxiliary target that depends on the model's representation.

Notation follows the local code: $\delta_s = (d_s^- - d_s^+)/(d_s^+ + \varepsilon)$, $\alpha_s = \delta_s/(\delta f + \varepsilon)$. Here d^+ and d^- are the Euclidean distances to the positive/negative center, f is the multimodal task, s a unimodal task and $\varepsilon=10^{-8}$. The new target is $\tilde{y}_s = \tfrac{1}{2}\alpha_s y + \tfrac{1}{2}(y + \delta_s - \delta f)$, clipped to $[-H,H]$ with $H=3$. For epoch n the update is $y_s \leftarrow [(n-1)/(n+1)]y_s + [2/(n+1)]\tilde{y}_s$. Updates start after the first epoch. [1; local trainer SELF_MM.py, update_labels]

The loss is the mean absolute error of the multimodal prediction plus the weighted absolute errors of the auxiliary tasks: $L = \text{mean}(|\hat{y}M - y|) + \Sigma_s \text{mean}(\tanh(|y_s - y|) \cdot |\hat{y}_s - y_s|)$. Auxiliary targets that differ more from the multimodal label therefore carry more weight. This weight in the training loss is not the same as the fusion weight tested at inference in this project. [1; local trainer, weighted_loss]

Implementation Limits

The original MMSA training loop was kept. With update_epochs=4, the gradients of four mini-batches are accumulated; the loss is not divided by this number. This setup cannot be presented as exactly equivalent to training with batch=16. Code review also shows that, with 1284 samples and batch=4, the last accumulation group of the 321 mini-batches is incomplete: mini-batch 320 produces the last step, and the gradient of the last mini-batch is zeroed at the start of the next epoch. This is an inherited implementation limitation noticed after the checkpoints were produced. A and B used the same loop; the code was not changed here, because the old results could then no longer be assumed to stay the same.

Checkpoint selection uses the validation Loss. This Loss is the mean of the batch L1 means; because the last batch of the 229 samples holds a single sample, it need not equal the equally

weighted MAE over all samples. The training data loader also uses shuffle=True for validation. The later fixed-checkpoint evaluations use shuffle=False and weight all samples equally. The MAE in the report is therefore kept explicitly separate from the Loss used for selection during training.

4 REPRODUCTION TARGET, IMPLEMENTATION AND DEVIATIONS

Target and Expected Observation

The numeric target in the 15 October plan was to approach Self-MM's MOSI performance, checking a band of ±1.5 percentage points for accuracy/F1 and ±0.03 for MAE. These bands are not a statistical significance test. Re-testing the paper's full claim that "the multi-task structure beats its competitors" would need matched competitor training and auxiliary-task removal experiments. These are not part of this five-seed package, so only the reproduction of the adapted baseline performance is reported here. The paper's causal ablation result is not counted as reproduced.

Table 1 Published results and our clean test baseline measurement

Source	MAE ↓	Non0 Acc % ↑	Non0 F1 % ↑	Corr ↑
Self-MM paper, Table 1 (reported by the authors)	0.713	85.98	85.95	0.798
MMSA results table (result-stat.md)	0.7080	85.46	85.43	0.7963
This work, A (measured; 5 seeds)	0.7240 ± 0.0146	83.90 ± 1.01	83.88 ± 0.92	0.7834 ± 0.0057
Difference: this work − paper	+0.0110	−2.08	−2.07	−0.0146

The paper row is the authors' mean over five runs; no SD has been invented for it here. Because MAE and Corr in the MMSA table are scaled by 100, they were converted as 70.80→0.7080 and 79.63→0.7963. Our row is the mean ± sample SD of five separate test measurements recorded at the end of training; the single-run SD=0 value in the CSV was not used as uncertainty. Paper values are read from research/configs/published_reference.json and MMSA values from results/result-stat.md [9].

The baseline MAE differs from the paper's by +0.0110, inside the plan's MAE band. The accuracy difference is −2.08 percentage points and the F1 difference −2.07 percentage points; both lie outside the plan's ±1.5 point band. It therefore cannot be said that "the paper's results were reproduced exactly". Differences in batch size, architecture settings, source version, evaluation and feature file were examined first; the gap is not attributed directly to a failure of the method.

Table 2 Deviations from the source and the initial plan

Topic	Actual implementation and valid interpretation
Source implementation	MMSA 2.2.1 was used instead of the authors' repository. The MOSI config currently visible in the authors' repository has batch 32, audio/video LSTM 32/64 and fusion dropout 0.1. The local values are 4, 16/32 and 0.0. It was not verified that this current config matches the paper's historical run. [8]
Hyperparameters	Local audio/video learning rate 0.005; BERT 0.00005; others 0.001. No claim is made that all settings are equal across sources. The resolved local configs of A and B match.
Dropout	Independent input-level audio/video dropout with p=0.2 was used instead of the representation-level dropout with p=0.3 in the initial plan. This change was made during implementation and is not presented as part of the initial plan.
Weighting	3×softmax was used instead of the softmax in the initial plan, so that equal weights are 1. In v1, τ=1 was fixed and no temperature sweep was run. In protocol v2 the τ sweep was run only on validation and for diagnosis; τ^* was chosen with a rule written before the results were seen.
Model and condition scope	Matched five-seed LMF/MulT training was not done; competing architectures appear only in the literature comparison. The planned K8 text corruption was not run in v1; it was added in protocol v2 as K8–K10 for diagnosis (Section 6).
Scope of claims	A low-resource adapted reproduction and a controlled intervention comparison are valid. A full repeat with the original settings, superiority over all competitors and generalization to all forms of modality loss were not shown.

5 PROPOSED IMPROVEMENT AND ITS RATIONALE

The initial plan hypothesized that the prediction of a corrupted modality could drift away from the others and that reducing its influence could help. The expectation was this: if the unimodal prediction of the corrupted modality moves away from the others, its deviation (d) grows, its weight falls and the fusion is less affected by the corruption; on clean input the predictions are close to each other, so the weights stay near 1 and performance is preserved. This expectation rested on the assumption that all three unimodal heads carry information. Modality dropout was proposed to expose the model to missing input during training. The report keeps this earlier rationale and states separately the implementation changes (p=0.2, input level and the factor 3). No retroactive pre-registration claim is made for these details.

For dropout, independent $b_A, b_V \sim \text{Bernoulli}(0.2)$ are drawn for each sample, and the whole corresponding audio/video input sequence is zeroed. The expected rate of dropping both is $0.2 \times 0.2 = 0.04$. Text and sequence lengths do not change. The temporary forward patch only runs when model.training is set; it is disabled during evaluation and removed at the end of the run. This

input corruption can also indirectly affect the training dynamics of the auxiliary targets; it is not a change to the fusion layer alone.

For the weighting, ŷT, ŷA, ŷV are taken from a first pass. $d_m = |\hat{y}_m - (\Sigma_{j \neq m} \hat{y}_j)/2|$ and $w_m = 3 \exp(-d_m/\tau)/\Sigma_j \exp(-d_j/\tau)$, with $\tau=1$. The three weights sum to 3. In a second pass, $h'_m=w_m h_m$ is applied and fed into the existing fusion layer. The weights are not 0–1 probabilities; they can exceed 1. A missing modality is not automatically given zero weight.

This method adds no newly learned parameters, but in the current implementation it needs two forward passes. It therefore cannot be said that "the computational cost is zero". In addition, the evaluation script runs one more pass for the equal-weight check. The total evaluation time including the checks is not the real inference latency of an application.

Table 3 Four methods that separate the interventions

Method	Checkpoint training	Inference fusion	Effect measured
A	Standard Self-MM	Original	Baseline
B	Input dropout in training only	Original	B−A dropout effect
C	Same checkpoint as A	$\tau=1$ weighted	C−A weighting effect
D	Same checkpoint as B	$\tau=1$ weighted	D−B weighting effect on top of dropout

Second Stage: Diagnostic Experiments That Separate the Causes of the Negative Result

When C/D increased the error in every condition in v1, a second experiment was designed to identify the cause instead of changing the method after looking at the test set. There were four possible explanations: (a) the size of the weights creates a representation distribution the fusion never saw in training; (b) the harm comes from rescaling the dominant text block; (c) the agreement signal carries no sample-specific information; (d) audio/video already matter little to the model, so there is no error to fix when these modalities are corrupted. A control was defined for each: a τ sweep, a variant with the text weight fixed, weights shuffled across samples, and an oracle that gives zero weight to the known missing modality. Text corruption (K8–K10) was added for the case where the dominant modality is corrupted. The hypotheses and thresholds were committed before the GPU run (Section 6, Tables 5–6).

This stage does not propose a new improvement; it aims to explain why the v1 intervention did not work. No new method was chosen, no retraining was done, and the test was run only once.

6 EXPERIMENTAL DESIGN, DATA SPLITS, METRICS AND RESOURCES

Data and Corruption Protocol

CMU-MOSI was used with train=1284, validation=229 and test=686 samples. On validation, the Non0 metrics cover 216 samples. The data splits come from the pre-extracted feature file; no new features were extracted from raw video. The literature uses speaker-independent evaluation for

MOSI; in this study the file hash was verified, but the speaker identities were not re-checked with an independent split audit. [1,2,7]

File: MOSI/Processed/unaligned_50.pkl. SHA-256: 78e0f8b5ef8ff71558e7307848fc1fa929ecb078203f565ab22b9daab2e02524. Text is BERT input; audio/video dimensions are 5/20; maximum sequence lengths are 50/500/375. Corruption is applied at the level of pre-extracted features, not to raw audio or video.

Table 4 Conditions K0–K10

Condition	Operation
K0	All modalities present
K1	Zero the whole audio input
K2	Zero the whole video input
K3	Zero the audio and video inputs together
K4	Zero valid audio/video time steps independently with probability 30%
K5	Same protocol at 50%; contains the K4 masks
K6	Same protocol at 70%; contains the K5 masks
K7	Zero-mean Gaussian noise on audio/video; std=0.5×train feature std
K8	Each valid text token becomes [UNK] independently with probability 30%; audio/video clean
K9	The same operation with probability 60%; contains the K8 masks
K10	All valid text tokens become [UNK]; audio/video clean

In K4–K6 all features of a selected time step are zeroed; length, padding and text are kept. The audio and video random generators are independent, and the masks are nested across rates. The K7 scales are computed only from the valid time steps in train; no noise is added to features with std=0. The training seeds are 1111–1115; the corruption seed is 1111 for all models. The inputs of the same condition are matched by hash, independently of the order of models and training seeds. K8–K10 are used only in protocol v2: [CLS] and the final [SEP] are kept; the attention mask, segments and padding do not change. This is the same missing-text rule MMSA uses for TFR-Net ([UNK]=100) [4,7]. The text masks are nested (K8 $\subset$ K9 $\subset$ K10); the randomness is derived from the protocol, split, sample and modality. The realized masking rates on test are 30.2%, 61.2% and 100%.

Diagnostic Protocol Version 2 and Pre-Registration

Because the negative result of v1 did not explain its cause, a second, inference-only protocol was written for the same ten checkpoints (research/selfmm_diagnostics_protocol_v2.md in the repository). Every condition K0–K10 is evaluated with the variants in Table 5; $d_m = |\hat{y}_m - \text{mean}(\hat{y}_{others})|$ is computed from the first-pass predictions of the unimodal heads. As in v1, the weights are applied to the text (768), audio (16) and video (32) blocks at the input of the first fusion layer.

Table 5 Protocol v2 weighting variants

Variant	Weight (text, audio, video)	Question it separates
plain (A/B)	No weighting	Reference
uniform	(1, 1, 1)	Identity check; must equal plain
agree τ	$3 \cdot \text{softmax}(-d/\tau)$, $\tau \in \{0.25; 0.5; 1; 2; 4; 8; 16\}$; $\tau=1$ is C/D in v1	Does the harm grow with the amount of rescaling?
Text fixed (agree_av)	Text 1; audio/video $2 \cdot \text{softmax}(-[d_A, d_V])$	Does the harm come from rescaling the text block?
Shuffled τ=1	$\tau=1$ weight vectors are shuffled across samples with a fixed permutation	Does the agreement signal carry sample-specific information?
Oracle	Known missing modality 0, the others 1 (K1, K2, K3, K10)	Would there be room to improve if the signal were perfect?

The hypotheses, decision criteria and expectations were written before the results were seen (Table 6). The criteria are applied on validation (229 samples), with the baseline checkpoint and over five seeds; the dropout checkpoint is reported as secondary. The decisions are descriptive; no p-values are computed.

Table 6 Pre-registered hypotheses and decision criteria

Code	Hypothesis	Support criterion (validation, baseline, five seeds)	Expected beforehand
H1	Harm grows with the amount of rescaling	Mean MAE over K0–K7 falls at every step as τ grows, and at K0 $\lvert\tau 16 - \text{plain}\rvert \le 0.01$	Supported
H2	Rescaling the text block is the main source of harm	At K0 (text fixed − plain) < 0.25 × (τ1 − plain), and text fixed < τ1 in 5/5 seeds	Supported
H3	The agreement signal carries useful sample-specific information	At K0 τ1 < shuffled in at least 4/5 seeds and mean difference ≤ −0.01	Not supported
H4	There is little room to improve under audio/visual corruption	$\lvert\text{oracle} - \text{plain}\rvert \le 0.01$ in each of K1, K2, K3	Supported
H5a	The model relies mainly on text	For plain, MAE(K10) − MAE(K0) ≥ 0.10	Supported
H5b	The rule helps when text is corrupted	At K9 or K10 τ1 < plain in at least 4/5 seeds and mean difference ≤ −0.01	Not supported

The order can be verified from the Git history. The protocol, script and criteria were committed on 25 October 2026 at 13:51 (e3e73b9); the first validation run started at 13:52. τ* is the τ with the lowest MAE on validation, averaged with equal weight over conditions K0–K10; the selection and the validation summary were committed at 15:50 (df50f85), and the test runs started at 15:51 and ran once per seed (339f2ed). No run counted as complete unless it reproduced the v1 A/B/C/D metrics on K0–K7 within a tolerance of 1e-5 (128 checks). The v1 validation and test results were known at that time; the H4 and H5a expectations rest partly on those observations.

Metrics and Uncertainty

MAE is the mean of |prediction−label|; lower is better. Non0 accuracy and support-weighted binary F1 evaluate the negative/positive distinction on samples whose true label is not zero. Pearson Corr measures linear co-variation. Because accuracy measures the class direction and MAE the sentiment intensity, one can stay the same while the other gets worse. The thresholds are consistent with the current MMSA implementation.

For each condition, the five-seed mean and the sample standard deviation are computed with $n-1=4$ in the denominator. A paired difference is first taken between two methods within the same seed; then the mean and SD of the five differences are computed. The SD is neither a confidence interval nor a standard error. Because the eight conditions use the same data and nested masks, they do not count as 40 independent repetitions. This report does not present p-values or claims of definite superiority as if a significance test had been done.

The seed SD shows training variability. For the uncertainty that comes from test sampling, the 686 test samples were resampled with replacement 2000 times; the same index sets were applied to all

seeds, methods and conditions, and a 95% percentile interval is given for the five-seed mean difference [10] (research/bootstrap_selfmm_diagnostics.py). The list of comparisons was fixed independently of the results. The bootstrap interval does not include seed variability, so the two are reported together. No multiple-comparison correction was applied.

Hardware, Training Settings and Cost

The experiments ran on Windows 11, Python 3.10.5 and an NVIDIA GeForce RTX 3050 Ti Laptop GPU with 4 GB. Torch 2.3.1+cu121, CUDA 12.1, MMSA 2.2.1, NumPy 1.26.4, Transformers 4.30.2, sentencepiece 0.2.2, pandas 1.5.3 and scikit-learn 1.3.2 were recorded. For Windows import compatibility, sentencepiece is imported before torch/MMSA. The training settings were num_workers=0, batch=4, Adam, early_stop=8 and update_epochs=4, with BERT fine-tuning enabled.

Table 7 Recorded training runs and resource measurements

Seed	Method	Epochs	Best epoch	MMSA total s	Peak allocated MiB	Peak reserved MiB
1111	A	11	3	No record	No record	No record
1111	B	14	6	402.36	2106.21	2378
1112	A	16	8	461.34	2106.21	2376
1112	B	20	12	568.72	2106.21	2378
1113	A	18	10	492.89	2106.21	2376
1113	B	22	14	618.75	2106.21	2378
1114	A	16	8	445.54	2106.21	2376
1114	B	17	9	482.34	2106.21	2378
1115	A	15	7	418.04	2106.21	2376
1115	B	19	11	535.43	2106.21	2378

In the matched 1112–1115 records, the total MMSA time is 454.46 ± 31.25 seconds for A and 551.31 ± 57.33 seconds for B (n=4). Because the seed 1111 baseline time/memory metadata is not in these files, no five-run resource mean has been invented. Peak allocated memory is 2106.21 MiB in the measured runs. Reserved memory is 2376 MiB for A and 2378 MiB for B; these values do not represent total system GPU use.

The time covers the whole MMSA_run call: loading, training, validation, the final test and cleanup. Because the B runs stopped after different numbers of epochs, the time difference cannot be read as the cost of the dropout operation alone. These records are kept in the repository as research/report_tables/training_resources.csv. C/D add no new parameters but need two forward passes; the controlled inference measurement is given in Section 7.

Source Version and Chronology of Test Access

The Git HEAD of the seed 1112 training records is bc67628ae90fcd85b49ef6a3f6202358f4b3872e; the HEAD of the 1113–1115 records is 53a366864719986d95865738e16309e08fde9c86. Because the actual file hashes in the working tree were recorded separately, HEAD alone is not a claim of a clean working tree. The commit order of protocol v2 is given above; the submitted version is pinned by the tag on the first page.

MMSA runs an automatic test on the selected checkpoint at the end of training; for seed 1111 there is also an earlier K0–K3 test analysis. The test set therefore cannot be presented as a completely unseen evaluation set. Validation was used for later method decisions; no new τ or dropout search should be made on the basis of the current test observations. The v1 A/B/C/D K0–K7 test comparison was completed with this fixed protocol; the protocol v2 test was run once, after the validation selection had been committed. Neither counts as a new independent holdout.

7 RESULTS, CONTROLS, UNCERTAINTY AND ERROR ANALYSIS

Five-Seed Validation Summary

Table 8 Clean validation K0, mean and sample SD

Method	MAE ↓	Non0 Acc % ↑	Non0 F1 ↑	Corr ↑
A	0.7168 ± 0.0202	84.91 ± 0.70	0.8487 ± 0.0065	0.7986 ± 0.0066
B	0.7049 ± 0.0117	85.56 ± 0.39	0.8553 ± 0.0034	0.8024 ± 0.0039
C	0.8665 ± 0.0513	84.72 ± 0.65	0.8468 ± 0.0061	0.7667 ± 0.0362
D	0.9293 ± 0.0667	85.37 ± 0.84	0.8535 ± 0.0080	0.7493 ± 0.0425

On clean validation, dropout B lowered the mean MAE relative to A by 0.01194; the SD of the paired difference is 0.02806. Four of the five seeds improved and seed 1114 got worse. Because the mean difference is smaller than the variability across seeds, no strong claim of superiority should be made from a single run. Weighting increased MAE in all five seeds for both C−A and D−B.

Table 9 Validation K0 paired differences

Comparison	MAE difference ± SD	Acc difference pp ± SD	Seeds with lower MAE
B−A	−0.0119 ± 0.0281	0.65 ± 0.53	4/5
C−A	0.1497 ± 0.0414	−0.19 ± 0.25	0/5
D−B	0.2244 ± 0.0609	−0.19 ± 0.70	0/5
D−A	0.2125 ± 0.0619	0.46 ± 1.09	0/5

Table 10 MAE mean and SD in all validation conditions

Condition	A	B	C	D
K0	0.7168 ± 0.0202	0.7049 ± 0.0117	0.8665 ± 0.0513	0.9293 ± 0.0667
K1	0.7170 ± 0.0201	0.7050 ± 0.0117	0.8659 ± 0.0516	0.9282 ± 0.0658
K2	0.7196 ± 0.0215	0.7051 ± 0.0118	0.8700 ± 0.0452	0.9205 ± 0.0566
K3	0.7197 ± 0.0214	0.7052 ± 0.0118	0.8693 ± 0.0455	0.9211 ± 0.0570
K4	0.7175 ± 0.0206	0.7048 ± 0.0117	0.8678 ± 0.0497	0.9289 ± 0.0661
K5	0.7177 ± 0.0209	0.7048 ± 0.0117	0.8679 ± 0.0488	0.9281 ± 0.0650
K6	0.7183 ± 0.0208	0.7049 ± 0.0117	0.8688 ± 0.0475	0.9268 ± 0.0631
K7	0.7168 ± 0.0200	0.7050 ± 0.0116	0.8655 ± 0.0520	0.9293 ± 0.0667

Weighting increased MAE relative to the matched base method in all five seeds in each of the eight conditions. This strong descriptive consistency concerns only the current $\tau=1$ rule. The fact that the accuracy/F1 of C and D sometimes stays the same does not offset the MAE harm. At K3, the MAE increase relative to each method's own K0 is about 0.002847 for A and 0.000267 for B. However, a small change under corruption can also appear in a model that makes little use of audio/video; on its own it is not strong evidence of multimodal robustness.

The mean effect of K7 noise is very small: the MAE change is about −0.000025 for A and +0.000116 for B. These values come from a single fixed corruption draw. No conclusion is drawn that noise helps or that all noise levels are harmless. No wider experiment over noise seeds or strengths was carried out for this report.

Test Results: A/B/C/D, K0–K10

Table 11 and Figure 1 give the five-seed test results of the v1 methods in conditions K0–K10. The K0–K7 values are identical to the v1 result package; K8–K10 were measured only in protocol v2. The clean test MAE of A is 0.7240 ± 0.0146. The Non0 accuracy table is in Appendix A.

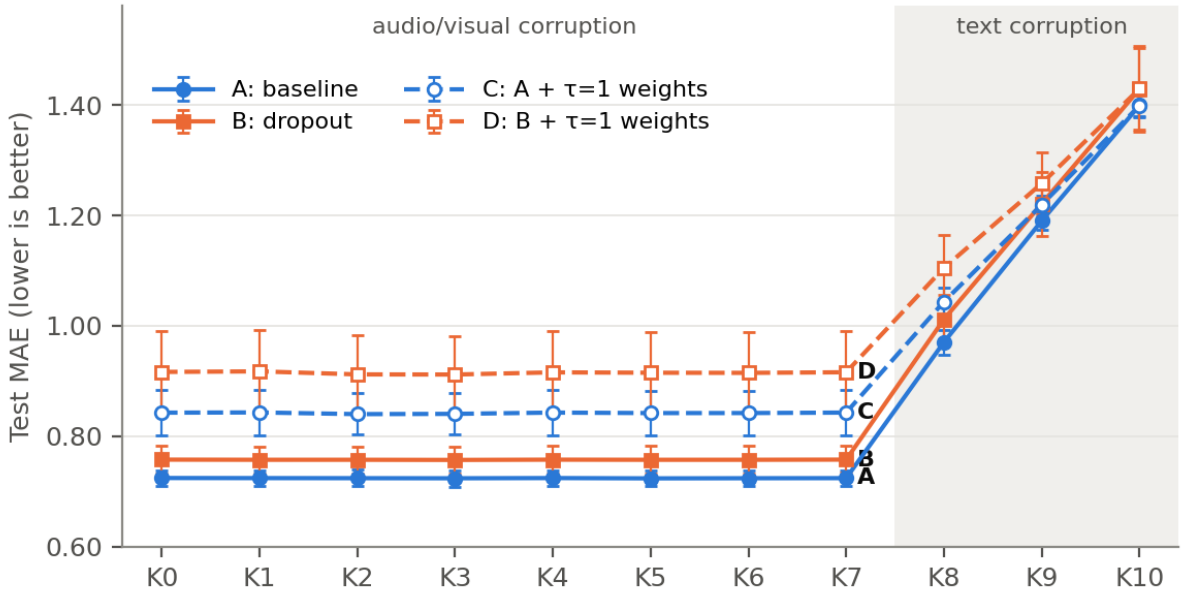


Figure 1 Test MAE by condition, A/B/C/D (five-seed mean ± seed SD). The gray area marks text corruption (K8–K10). C and D apply τ=1 weights to the same checkpoints as A and B.

Table 11 Test MAE by condition, A/B/C/D (mean ± sample SD, five seeds)

Condition	A	B	C	D
K0	0.7240 ± 0.0146	0.7574 ± 0.0244	0.8424 ± 0.0412	0.9163 ± 0.0747
K1	0.7239 ± 0.0146	0.7571 ± 0.0246	0.8428 ± 0.0410	0.9173 ± 0.0758
K2	0.7237 ± 0.0150	0.7571 ± 0.0244	0.8398 ± 0.0375	0.9118 ± 0.0709
K3	0.7235 ± 0.0150	0.7568 ± 0.0246	0.8403 ± 0.0374	0.9115 ± 0.0703
K4	0.7239 ± 0.0143	0.7573 ± 0.0245	0.8426 ± 0.0410	0.9155 ± 0.0743
K5	0.7234 ± 0.0141	0.7570 ± 0.0247	0.8417 ± 0.0410	0.9148 ± 0.0742
K6	0.7236 ± 0.0142	0.7571 ± 0.0248	0.8418 ± 0.0404	0.9146 ± 0.0738
K7	0.7238 ± 0.0145	0.7574 ± 0.0244	0.8426 ± 0.0415	0.9157 ± 0.0740
K8	0.9703 ± 0.0223	1.0117 ± 0.0445	1.0436 ± 0.0259	1.1047 ± 0.0594
K9	1.1903 ± 0.0167	1.2212 ± 0.0582	1.2193 ± 0.0154	1.2579 ± 0.0559
K10	1.3994 ± 0.0199	1.4291 ± 0.0773	1.3990 ± 0.0208	1.4294 ± 0.0736

The small validation gain of dropout did not carry over to test: at K0, B−A = +0.0334 [+0.0166; +0.0499] and B is worse in four of the five seeds; the Non0 accuracy difference is −1.01 pp [−2.09; +0.12] (Table 12). No general benefit is therefore claimed for dropout.

τ=1 weighting increased MAE in every K0–K7 condition and every seed: at K0, C−A = +0.1184 [+0.0909; +0.1475] and D−B = +0.1589 [+0.1283; +0.1930]. The change in Non0 accuracy is

small (C−A −0.09 pp [−0.25; +0.06]); the rule largely keeps the sentiment direction but damages the intensity prediction.

Table 12 Paired test differences: point estimate, 95% bootstrap interval and seed SD

Difference	Condition	MAE difference [95% CI]	Seed SD	Seeds with lower MAE	Non0 Acc difference pp [95% CI]
B−A	K0	+0.0334 [+0.0166; +0.0499]	0.0364	1/5	−1.01 [−2.09; +0.12]
B−A	K3	+0.0333 [+0.0165; +0.0496]	0.0367	1/5	−1.10 [−2.10; −0.06]
B−A	K7	+0.0336 [+0.0167; +0.0502]	0.0364	1/5	−0.98 [−2.03; +0.12]
B−A	K9	+0.0309 [+0.0189; +0.0426]	0.0719	2/5	−2.74 [−4.30; −1.13]
B−A	K10	+0.0297 [+0.0226; +0.0366]	0.0926	4/5	−0.49 [−2.11; +1.12]
C−A	K0	+0.1184 [+0.0909; +0.1475]	0.0416	0/5	−0.09 [−0.25; +0.06]
C−A	K3	+0.1167 [+0.0896; +0.1466]	0.0398	0/5	−0.06 [−0.19; +0.06]
C−A	K7	+0.1188 [+0.0911; +0.1482]	0.0419	0/5	−0.03 [−0.19; +0.12]
C−A	K9	+0.0291 [+0.0180; +0.0404]	0.0062	0/5	−0.06 [−0.48; +0.34]
C−A	K10	−0.0004 [−0.0010; +0.0002]	0.0019	2/5	−0.06 [−0.21; +0.09]
D−B	K0	+0.1589 [+0.1283; +0.1930]	0.0616	0/5	+0.00 [−0.09; +0.09]
D−B	K3	+0.1547 [+0.1240; +0.1885]	0.0559	0/5	+0.06 [+0.00; +0.15]

Difference	Condition	MAE difference [95% CI]	Seed SD	Seeds with lower MAE	Non0 Acc difference pp [95% CI]
D−B	K7	+0.1583 [+0.1280; +0.1923]	0.0609	0/5	−0.06 [−0.18; +0.00]
D−B	K9	+0.0367 [+0.0243; +0.0491]	0.0167	0/5	+0.18 [−0.03; +0.40]
D−B	K10	+0.0004 [−0.0007; +0.0014]	0.0043	1/5	−0.09 [−0.33; +0.15]

The MAE of A does not change even when audio and video are removed completely: K3 − K0 = −0.0005 [−0.0033; +0.0024]. When text is corrupted with [UNK], however, the error rises sharply: K8 +0.246, K9 +0.466, K10 +0.675; at K10 Non0 accuracy falls from 83.90% to 55.06%. Our interpretation is that insensitivity to audio/visual corruption is not robustness but a sign that the model hardly uses these modalities; this interpretation is tested with the oracle experiment below.

Results of the Diagnostic Experiments (Protocol v2)

Table 13 gives the six pre-registered decisions. All six matched the expectation and replicated on test. Unless stated otherwise, the numbers below are five-seed test means for the baseline checkpoint; the validation numbers are in the summary package in the repository.

Table 13 Hypothesis decisions: validation and test

Code	Hypothesis	Expected beforehand	Validation decision	Test (replication)
H1	Harm grows with the amount of rescaling (τ sweep)	Supported	Supported	Supported
H2	Rescaling the text block is the main source of harm	Supported	Supported	Supported
H3	The agreement signal carries useful per-example information	Not supported	Not supported	Not supported
H4	Little room to improve under audio/visual corruption (oracle)	Supported	Supported	Supported
H5a	The model relies mainly on text	Supported	Supported	Supported
H5b	The rule helps when text is corrupted	Not supported	Not supported	Not supported

H1, dose–response: the harm falls steadily as τ grows (Figure 2, Table 14). The mean MAE over K0–K7 on validation drops from 1.2300 at τ=0.25 to 0.7216 at τ=16; unweighted fusion gives 0.7179. The test curve shows the same pattern (1.2052 → 0.7238; unweighted 0.7237). The τ* =

16 selected on validation lies at the edge of the grid and gives the same result as unweighted fusion on test: K0 difference +0.0002 [−0.0019; +0.0023]. In this family of rules, the best option is effectively not to weight at all.

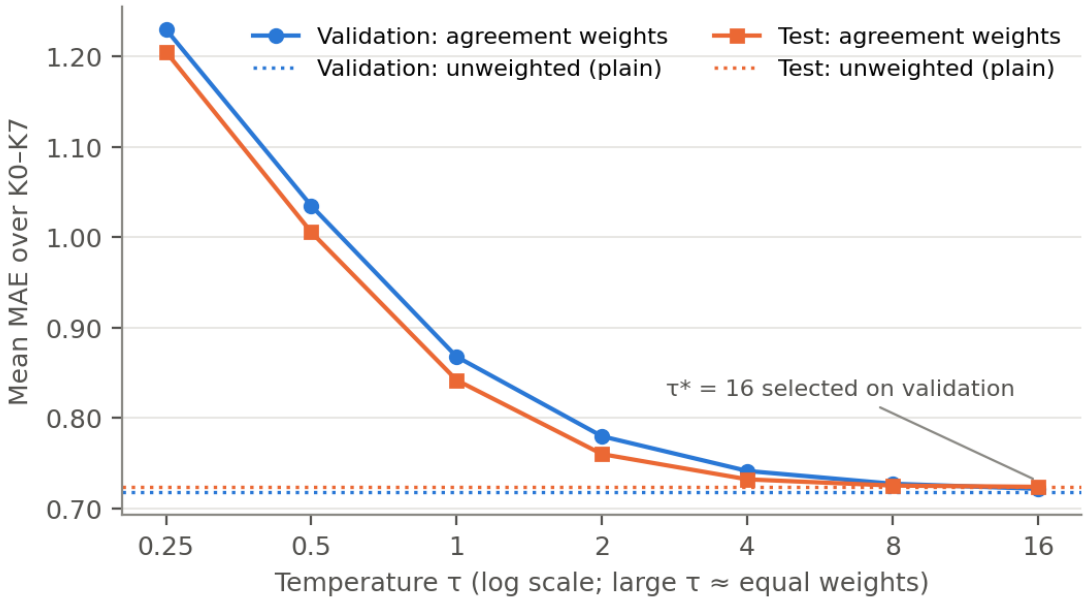


Figure 2 τ sweep: mean MAE over K0–K7 with the baseline checkpoint. The dotted lines show unweighted fusion. As τ grows, the weights approach 1.

Table 14 τ sweep, baseline, mean MAE over K0–K7

Split	plain	τ=0.25	τ=0.5	τ=1	τ=2	τ=4	τ=8	τ=16
Validation	0.7179	1.2300	1.0347	0.8677	0.7797	0.7413	0.7272	0.7216
Test	0.7237	1.2052	1.0055	0.8417	0.7600	0.7320	0.7252	0.7238

H2, text block: when the text weight is held at 1 and only audio/video are weighted, the harm disappears: at K0 the difference is +0.0005 on validation and −0.0004 [−0.0011; +0.0003] on test, while the same difference for the τ=1 rule is +0.1184 (Table 15, Figure 3). The harm comes from rescaling the dominant 768-dimensional text representation; the weights of the audio and video blocks hardly change the result.

H3, signal content: the agreement signal is even worse than the same weights randomly shuffled across samples: at K0, τ=1 − shuffled = +0.0404 [+0.0225; +0.0596], and τ=1 is not better in any seed. The rule carries no useful sample-specific information; on the contrary, it is more harmful than random weights.

Table 15 Test MAE differences of the diagnostic variants (baseline)

Comparison	Condition	MAE difference [95% CI]	Seeds with lower MAE
τ=1 (C) − plain	K0	+0.1184 [+0.0909; +0.1475]	0/5
τ=1 (C) − plain	K3	+0.1167 [+0.0896; +0.1466]	0/5
τ=1 (C) − plain	K10	−0.0004 [−0.0010; +0.0002]	2/5
τ=16 − plain	K0	+0.0002 [−0.0019; +0.0023]	2/5
τ=16 − plain	K3	−0.0003 [−0.0024; +0.0020]	3/5
τ=16 − plain	K10	−0.0000 [−0.0001; +0.0000]	2/5
Text fixed, A/V weighted − plain	K0	−0.0004 [−0.0011; +0.0003]	4/5
Text fixed, A/V weighted − plain	K3	−0.0002 [−0.0003; −0.0001]	3/5
Text fixed, A/V weighted − plain	K10	−0.0004 [−0.0008; +0.0001]	3/5
τ=1 − shuffled τ=1	K0	+0.0404 [+0.0225; +0.0596]	0/5
τ=1 − shuffled τ=1	K3	+0.0451 [+0.0279; +0.0623]	0/5
τ=1 − shuffled τ=1	K10	+0.0003 [−0.0003; +0.0009]	1/5
Oracle − plain	K1	−0.0007 [−0.0016; +0.0002]	3/5
Oracle − plain	K2	−0.0002 [−0.0006; +0.0001]	4/5
Oracle − plain	K3	−0.0009 [−0.0021; +0.0003]	4/5
Oracle − plain	K10	+0.0224 [+0.0098; +0.0353]	2/5
plain K8 − plain K0	K8	+0.2463 [+0.1974; +0.3018]	0/5
plain K9 − plain K0	K9	+0.4662 [+0.4051; +0.5293]	0/5
plain K10 − plain K0	K10	+0.6754 [+0.6100; +0.7490]	0/5

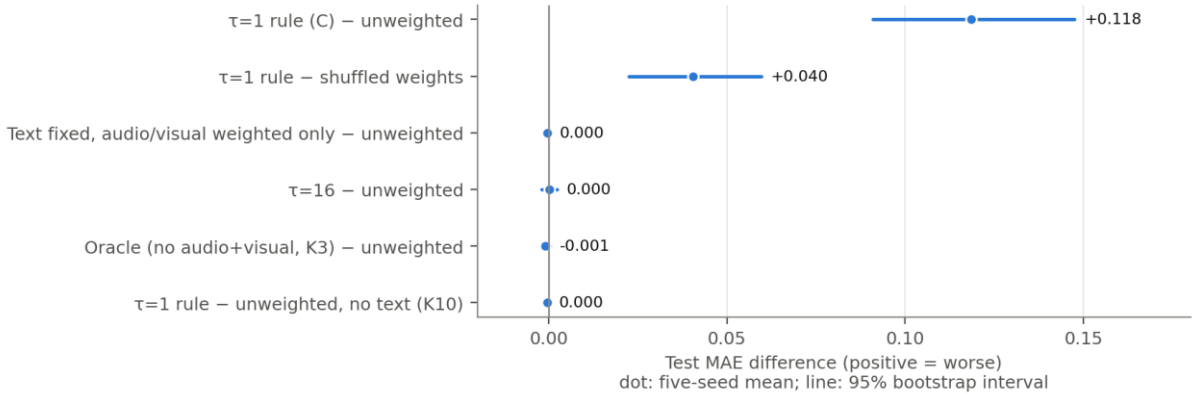


Figure 3 Test MAE difference of the diagnostic variants against unweighted fusion. The dot is the five-seed mean and the line the 95% paired bootstrap interval; positive values are worse.

H4, room to improve: the oracle, which gives zero weight to the known missing modality, does not change MAE under audio/visual corruption (K1 −0.0007, K2 −0.0002, K3 −0.0009; all three intervals include zero). There is no error to fix in these conditions; a research question built on audio/visual corruption had limited room for improvement from the start.

H5, text corruption: the error grows when text is removed (H5a), but the rule does not help in this case either (H5b): at K9 τ=1 − plain = +0.0291 [+0.0180; +0.0404] (0/5 seeds), at K10 −0.0004 [−0.0010; +0.0002] (2/5 seeds). At K10 the oracle that gives zero weight to text is better than plain on validation (1.430 vs 1.464) but worse on test (1.422 vs 1.399; difference +0.0224 [+0.0098; +0.0353]). There was no prior criterion for this comparison, and its direction is inconsistent across splits.

The agreement between the decisions and the expectations is meaningful because the protocol was written before the results were seen. Still, because the H4 and H5a expectations rest partly on v1 observations, they are not presented as independent discoveries.

Mechanism: Why Does the Rule Suppress the Most Informative Modality?

The agreement rule treats the closeness of the unimodal heads' predictions as an indicator of reliability; for that, the heads must carry information. On test (baseline, K0), the predictions of the text head vary across samples with a standard deviation of 0.64–1.00, while the corresponding SDs of the audio head lie in 0.001–0.17 and those of the video head in 0.004–0.44 (Table 16). In most seeds the audio head produces an almost constant value.

Table 16 Spread of the unimodal heads across samples (baseline, K0, test)

Head	Prediction SD range (5 seeds)
Text	0.639 – 1.001
Audio	0.001 – 0.171
Visual	0.004 – 0.436

As a result, d_T grows as the text prediction moves away from neutral, and the rule suppresses strong text predictions: the mean text weight is 1.01 for samples with $|\hat{y}_T| < 0.5$ but 0.44 for those with $|\hat{y}_T| \geq 2$ (Figure 4a; the last bin contains 67 samples from three seeds). When text is

corrupted, the text prediction moves toward neutral, so the text weight instead increases: 0.783 at K0 and 0.994 at K10 (Table 17, Figure 4b). The rule penalizes the most informative modality, not the corrupted one.

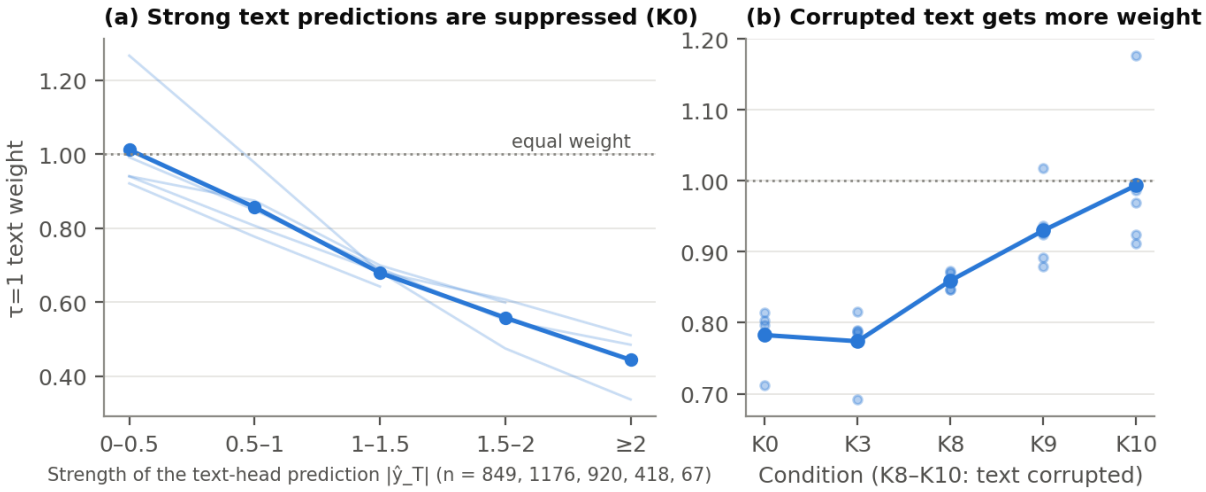


Figure 4 Mechanism of the agreement rule (baseline, test). (a) τ=1 text weight by the prediction strength of the text head at K0; (b) mean text weight by condition. Thin lines and small dots = seeds.

Table 17 Mean text weight under the τ=1 rule (baseline, test)

Condition	Mean ± seed SD	Seed range
K0	0.783 ± 0.041	0.712 – 0.814
K3	0.774 ± 0.047	0.692 – 0.815
K8	0.859 ± 0.012	0.847 – 0.873
K9	0.930 ± 0.054	0.879 – 1.018
K10	0.994 ± 0.107	0.912 – 1.177

Helped and Harmed Examples

Table 18 Extreme examples, seed 1111 clean validation

Comparison	Index	True label	Before	After	Absolute error difference
C−A	222	−2.60	−2.4840	−1.4546	1.0294
C−A	54	−1.40	−2.5672	−1.5455	−1.0216
D−B	77	2.60	2.5369	1.1558	1.3812
D−B	140	1.00	2.5618	1.2038	−1.3580

D−B, index 77: the true label is 2.60. The text prediction is 1.724, while the audio/video predictions are about −0.322/−0.393. Because audio and video are close to each other, the rule lowers the text weight to 0.450; the audio/video weights become 1.344/1.207. The fused prediction moves from 2.537 to 1.156 and the error increases. Here, “prediction agreement” did not pick the right modality.

D−B, index 140: the true label is 1.00. In the same way the prediction shrinks, from 2.562 to 1.204, so the error decreases. The same kind of rescaling can therefore help a sample that was over-predicted while harming a correct sample with strong sentiment. In both examples the sentiment direction stays positive; MAE changes without any change in accuracy. These are extreme examples selected by effect size, not a random or representative sample. No semantic check of the cause on the raw video/transcript has been done yet.

These examples are consistent with the mechanism isolated by the protocol v2 controls: most of the harm comes from rescaling the text block (H2), and the agreement signal is no better than randomly shuffled weights (H3). Whether retraining with the weighting would remove the shift in the representation distribution seen by the fusion has not been tested.

Quality and Resource Cost

Table 19 Inference cost (seed 1111, K0, test, batch 4, RTX 3050 Ti, 6 rounds)

Method	ms/example	examples/s	Peak allocated MiB	Forward passes
A	4.983 ± 0.026	200.7	436.54	1
B	5.073 ± 0.048	197.1	437.98	1
C	10.193 ± 0.057	98.1	436.54	2
D	10.350 ± 0.050	96.6	437.99	2

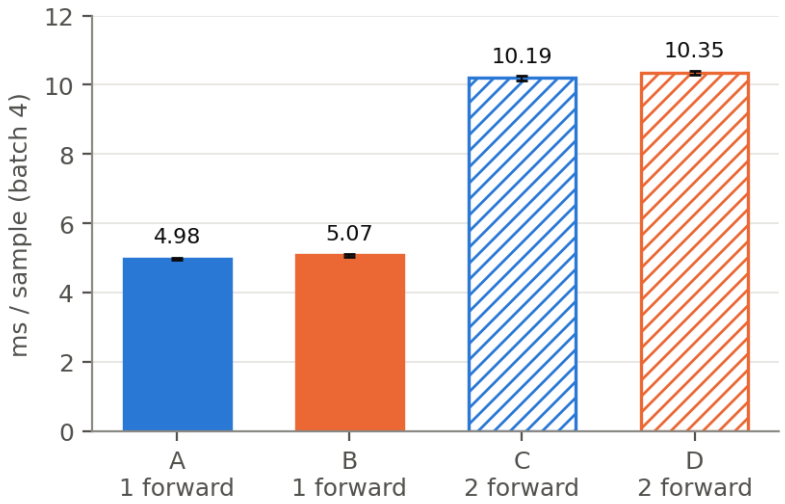


Figure 5 Inference time per sample (amortized over batches of 4; error bar = SD over rounds).

According to Table 19 and Figure 5, C/D take about twice as long per sample because of the two forward passes (C/A 2.045 ± 0.016; D/B 2.040 ± 0.013); the peak allocated memory difference is

6.5 KiB. The time is amortized over batches of 4; it is not batch-1 latency. Because the weighting lowers quality while doubling inference cost, it is not supported from a quality–cost point of view either. The training cost is in Table 7.

Controls and Data Integrity

In the five-seed analysis, 32 metric rows, 3664 prediction rows and 16 check rows were verified for each seed. The prediction rows are two checkpoints × eight conditions × 229 samples; because each row carries both the weighted and the unweighted outputs, they are not multiplied again by the four methods. It was checked that equal weights keep the original output, that the hook is removed even after an error, that the labels match, that the numeric values are finite and that the shared inputs are identical.

The aggregate analysis, keeping its 24 source files unchanged, produced 160 per-seed metric rows, 32 summaries, 160 paired differences, 32 paired summaries and 32 rows of change relative to K0. The source_files_unchanged, shared_inputs_verified, labels_matched and metrics_recomputed records are True. While this report was being prepared, the hashes of the 24 sources and six analysis outputs were checked again. No model or dataset was reloaded, and no training or checkpoint was changed.

Each of the ten protocol v2 runs (five seeds × validation/test) reproduced the v1 A/B/C/D metrics on K0–K7 in 128/128 checks (largest difference about 1e-15). With all weights at 1 the output stayed exactly the same; both checkpoints and all five seeds received the same corrupted inputs; on test the corrupted-input digests matched the v1 references 16/16. In total, 590 MAE values were recomputed from the compressed test predictions and matched the summary package. The summary packages, the bootstrap and the report tables can be re-verified without a GPU (research/README.md).

8 LIMITATIONS, CONCLUSIONS AND OPEN QUESTIONS

Main result: The adapted Self-MM baseline was run with five seeds; its clean test MAE is close to the paper's (0.7240 vs 0.713), while Non0 accuracy is 2.08 percentage points lower. τ=1 agreement weighting increased MAE in every K0–K7 condition and every seed and doubled inference time; the validation gain of dropout did not carry over to test. The pre-registered diagnostic experiments explained the negative result: the harm comes from rescaling the dominant text block; the agreement signal is no better than randomly shuffled weights; and because the audio and video heads vary very little across samples, the rule suppresses strong text predictions and gives more weight to text when it is corrupted. This rule therefore cannot be used as a reliability signal, and on MOSI, insensitivity to audio/visual corruption is not evidence of robustness.

Limitations: a single dataset (CMU-MOSI) with pre-extracted features; a single corruption seed and synthetic corruption at the feature level only; a small validation set (229 samples); five training repetitions; deviations from the paper's settings (batch 4, the MMSA implementation, the inherited gradient accumulation and checkpoint selection); a test set that had been seen before; all variants applied after training, so retraining with the weighting was not tested; the selected $\tau^* = 16$ lying at the edge of the grid; the inconsistent oracle result at K10 across splits; and no matched training of competing architectures. The bootstrap intervals reflect test-sampling uncertainty and the seed SD reflects training variability; neither guarantees generalization to other datasets.

Without stating these limits, terms such as “reliable”, “calibrated” or “generalizable” should not be used.

Open questions: Would training together with the weighting remove the shift in the representation distribution seen by the fusion? Why are the audio and video heads so weak on MOSI, and would stronger audio/visual features change the behavior of the same rule? How would learned uncertainty [6] or approaches that correct the dominant modality [5], used instead of agreement between predictions, behave under the same diagnostic protocol? Would different corruption seeds or real sensor faults change the result? These questions call for future experiments; they are not a reason to choose new settings on the test set.

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APPENDIX A SUPPLEMENTARY METRICS

Test Non0 accuracy %, K0–K10, mean ± sample SD, five seeds

Condition	A	B	C	D
K0	83.90 ± 1.01	82.90 ± 0.79	83.81 ± 0.90	82.90 ± 0.82
K1	83.90 ± 1.01	82.90 ± 0.79	83.81 ± 0.90	82.90 ± 0.82
K2	83.84 ± 1.14	82.74 ± 0.94	83.75 ± 1.05	82.84 ± 0.97
K3	83.84 ± 1.14	82.74 ± 0.94	83.78 ± 1.05	82.80 ± 0.93
K4	83.84 ± 1.07	82.80 ± 0.84	83.81 ± 0.86	82.80 ± 0.84
K5	83.96 ± 1.00	82.77 ± 0.89	83.84 ± 0.99	82.80 ± 0.93
K6	83.96 ± 1.02	82.77 ± 0.89	83.93 ± 0.97	82.80 ± 0.93
K7	83.87 ± 0.94	82.90 ± 0.79	83.84 ± 0.92	82.84 ± 0.83
K8	75.37 ± 1.16	73.35 ± 2.53	75.18 ± 1.22	73.48 ± 2.46
K9	66.68 ± 1.11	63.93 ± 4.51	66.62 ± 1.08	64.12 ± 4.66
K10	55.06 ± 3.82	54.57 ± 7.08	55.00 ± 3.94	54.48 ± 7.00

Validation Non0 accuracy %, K0–K7, mean ± sample SD, five seeds

Condition	A	B	C	D
K0	84.91 ± 0.70	85.56 ± 0.39	84.72 ± 0.65	85.37 ± 0.84
K1	84.91 ± 0.70	85.65 ± 0.33	84.72 ± 0.65	85.46 ± 0.84
K2	85.19 ± 0.93	85.19 ± 0.57	85.09 ± 0.95	85.19 ± 0.57
K3	85.19 ± 0.93	85.28 ± 0.60	85.09 ± 0.95	85.28 ± 0.60
K4	85.00 ± 0.84	85.46 ± 0.41	84.81 ± 0.76	85.28 ± 0.69
K5	85.09 ± 1.06	85.28 ± 0.69	85.09 ± 0.76	85.28 ± 0.69
K6	85.19 ± 0.93	85.37 ± 0.70	85.09 ± 0.95	85.37 ± 0.70
K7	84.81 ± 0.60	85.65 ± 0.57	84.72 ± 0.65	85.37 ± 0.84

Validation Non0 F1, K0–K7, mean ± sample SD, five seeds

Condition	A	B	C	D
K0	0.8487 ± 0.0065	0.8553 ± 0.0034	0.8468 ± 0.0061	0.8535 ± 0.0080
K1	0.8487 ± 0.0065	0.8562 ± 0.0031	0.8468 ± 0.0061	0.8544 ± 0.0081
K2	0.8516 ± 0.0089	0.8517 ± 0.0053	0.8507 ± 0.0090	0.8517 ± 0.0053
K3	0.8516 ± 0.0089	0.8526 ± 0.0059	0.8507 ± 0.0090	0.8526 ± 0.0059
K4	0.8496 ± 0.0077	0.8544 ± 0.0037	0.8477 ± 0.0069	0.8526 ± 0.0065
K5	0.8506 ± 0.0102	0.8526 ± 0.0065	0.8505 ± 0.0068	0.8526 ± 0.0065
K6	0.8515 ± 0.0089	0.8535 ± 0.0068	0.8506 ± 0.0089	0.8535 ± 0.0068
K7	0.8477 ± 0.0052	0.8562 ± 0.0053	0.8468 ± 0.0056	0.8535 ± 0.0080

Validation Pearson correlation, K0–K7, mean ± sample SD, five seeds

Condition	A	B	C	D
K0	0.7986 ± 0.0066	0.8024 ± 0.0039	0.7667 ± 0.0362	0.7493 ± 0.0425
K1	0.7986 ± 0.0068	0.8025 ± 0.0039	0.7671 ± 0.0367	0.7530 ± 0.0382
K2	0.7980 ± 0.0067	0.8025 ± 0.0036	0.7737 ± 0.0225	0.7606 ± 0.0266
K3	0.7980 ± 0.0069	0.8025 ± 0.0036	0.7742 ± 0.0229	0.7643 ± 0.0223
K4	0.7983 ± 0.0067	0.8025 ± 0.0038	0.7679 ± 0.0339	0.7509 ± 0.0403
K5	0.7983 ± 0.0068	0.8025 ± 0.0037	0.7696 ± 0.0313	0.7533 ± 0.0365
K6	0.7983 ± 0.0068	0.8025 ± 0.0037	0.7713 ± 0.0290	0.7560 ± 0.0333
K7	0.7986 ± 0.0066	0.8024 ± 0.0039	0.7671 ± 0.0363	0.7490 ± 0.0428

All report tables can be read from research/report_tables/tables_en.md, and the full bootstrap table of the 74 comparisons from research/diagnostics_v2/bootstrap/bootstrap_test.csv. Per-seed metrics for all variants and conditions of protocol v2 are in the validation and test packages under research/diagnostics_v2/.

APPENDIX B AI ASSISTANCE AND STUDENT RESPONSIBILITY

Tools used and their contributions: (1) ChatGPT/Codex (OpenAI) was used in the v1 stage for drafting the training and evaluation scripts, debugging, checking the experiment records, producing the first tables and editing the text of the 18 October report draft. (2) Exa was used to find sources and to access the primary paper texts. (3) Claude (Anthropic, Claude Code) was used

from 25 October 2026 for: a project audit against the brief and a list of gaps; writing the protocol v2 document and the diagnostic script (diagnose_selfmm_weighting.py); the five-seed summary, prediction export, bootstrap, report table and figure scripts; updating the repository documentation; drafting the text of Sections 1 and 4–8 and Appendices A–C in this version of the report and editing the Word document with python-docx; and translating the report and the presentation from the student's Turkish working version into English. Claude ran the summary, bootstrap, table and figure generation that does not need a GPU; these outputs can be regenerated with the verification commands in the repository.

What the student did: chose the topic and papers; carried out all GPU training and evaluation runs (v1 training, protocol v2 validation and test runs) on the student's own computer; approved and committed the protocol v2 thresholds before the GPU run; carried out all Git operations (commit, push, tag); compared the paper scores with Table 1 of the paper; and read and edited the report text and checked the numbers against the repository files.

No experimental result, measurement or source verification has been fabricated; steps that were not carried out are stated explicitly in the report.

APPENDIX C REQUIREMENTS OF THE BRIEF AND WHERE THE REPORT ADDRESSES THEM

Requirement	Place in the report	Evidence in the repository
Research question, scope, motivation and contributions	Section 1	YZM5110_PROJECT.md
Search approach; at least five papers; seven-field comparison matrix	Section 2	Six primary papers
Technical description of the chosen method	Section 3	src/MMSA/models/multiTask/SELF_MM.py, src/MMSA/trains/multiTask/SELF_MM.py
Testable target; published and measured results given separately; deviations	Section 4, Tables 1–2	research/configs/published_reference.json, results/result-stat.md
Source version, dependencies, data version, preprocessing, seeds, configs and commands	Section 6	research/README.md, requirements-research.txt, environment-observed.txt, configs/
Motivated intervention and an expectation written before the results	Section 5, Table 6	selfmm_diagnostics_protocol_v2.md (commit e3e73b9)
Baseline comparison under matched conditions; controls and ablation	Tables 3, 5, 11–15	A/B/C/D; uniform, τ sweep, text fixed, shuffled, oracle

Requirement	Place in the report	Evidence in the repository
Clean and corrupted conditions	Tables 4, 11; Figure 1	K0–K10; selfmm_corruptions.py, diagnose_selfmm_weighting.py
Task quality and a suitable resource measure	Tables 7, 19; Figure 5	results_package_v1/inference, report_tables/training_resources.csv
Uncertainty, study of the negative result, error analysis	Section 7, Tables 12–18, Figures 3–4	diagnostics_v2/ (validation, evaluation_test, bootstrap)
No tuning on the test set	Section 6, chronology	τ selected on validation only (df50f85); test run once (339f2ed)
Limitations, conclusions and open questions	Section 8	DIAGNOSTICS_V2.md
Public repository: setup, data access, commands, saved outputs, table and figure scripts	First page	research/README.md, make_selfmm_report_tables.py, make_selfmm_figures.py
Small tests and results that need extra resources are identified	Section 7, Controls	verify_results_package.py --check, diagnose_selfmm_weighting.py --self-test, --verify commands
Submitted version pinned by a commit or tag	First page	tag v1.0-submission
Statement of AI use	Appendix B	—

This mapping is meant to help the reader find where each requirement is addressed. The brief does not require a particular number of training seeds, LMF/MulT training, text corruption or every resource metric to be reported together; deviations from the student's initial plan are explained in Table 2.